

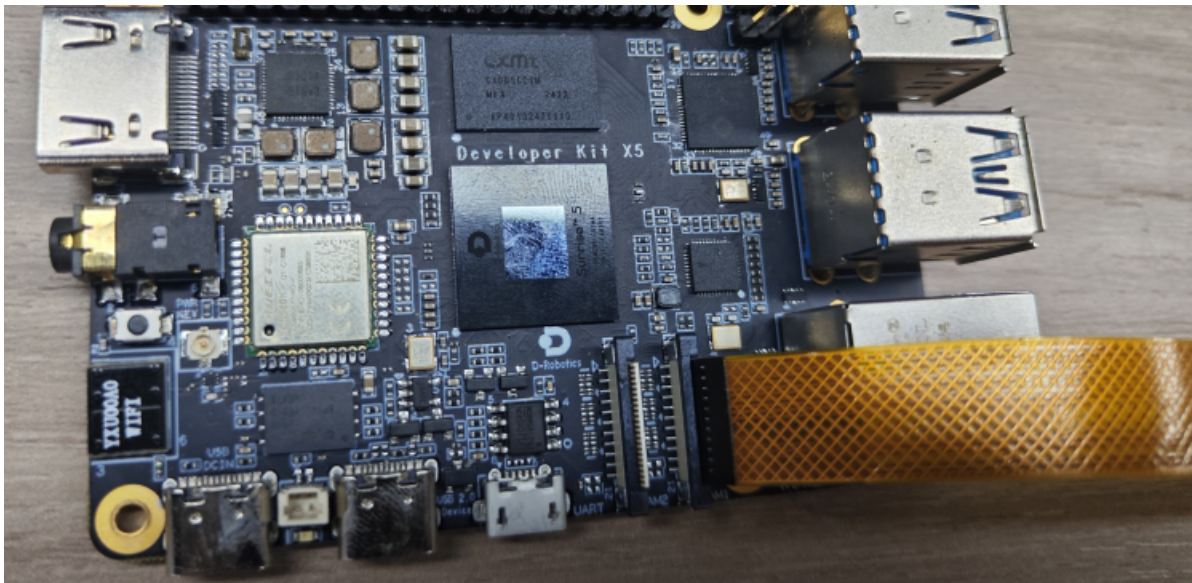
# Target Detection Algorithm - YOLOv5

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## 1. Environment Preparation

This example is based on the `YOLOv5` model and implements a camera target detection algorithm function. Users can preview the detection results through a monitor.



- With the development board powered off, connect the `MIPI` camera to the development board
- Connect the development board and the monitor with an HDMI cable
- Power on the development board and log in through the command line

## 2. How to Run

To run this case, you need to turn off the screen display first

```
sudo systemctl stop lightdm
```

After turning off the screen, perform the following operations:

Place the `01_mipi_camera_yolov5s.py` program from the source code summary into (replace it in) the `/app/pydev_demo/08_mipi_camera_sample` path on the RDK X5.

```
cd /app/pydev_demo/08_mipi_camera_sample
sudo python3 ./01_mipi_camera_yolov5s.py --model-path
/app/model/basic/yolov5s_672x672_nv12.bin
```

### Parameter configuration:

- --model-path : Model path

If you want to turn the screen back on, enter `sudo systemctl start lightdm`

## 3. Expected Results

After the program runs correctly, it will output the video and the rendered image after algorithm detection through the `HDMI` interface. Users can preview it through the monitor.

The running log is as follows:

```
set camera fps: 30,width: -1,height: -1
Camera 0:
    mipi_host: 0
Camera 1:
    mipi_host: 2
Camera 2:
    mipi_host: 0
Camera 3:
    mipi_host: 0
mipi mclk is not configed.
Searching camera sensor on device: /proc/device-tree/soc/cam/vcon@0 i2c bus: 6
mipi rx phy: 0
WARN: Sensor Name: sc1330t, Expected Chip ID: 0xCA18, Actual Chip ID Read: 0x00
WARN: Sensor Name: irs2875-tof, Expected Chip ID: 0x2875, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc230ai-10fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc230ai-10fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc230ai-10fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc230ai-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc230ai-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc230ai-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-1280p, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-1280p, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-1280p, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-1280p, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-1280p, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-hdr-2lane, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-hdr-2lane, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-hdr-2lane, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
WARN: Sensor Name: sc132gs-hdr-2lane, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00
```

WARN: Sensor Name: sc132gs-hdr-2lane, Expected Chip ID: 0x132, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc035hgs, Expected Chip ID: 0x31, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc035hgs\_mono, Expected Chip ID: 0x31, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc035hgs\_mono, Expected Chip ID: 0x31, Actual Chip ID Read: 0x00

WARN: Sensor Name: ov5640, Expected Chip ID: 0x5640, Actual Chip ID Read: 0x00

WARN: Sensor Name: f37, Expected Chip ID: 0xF37, Actual Chip ID Read: 0x00

[0] INFO: Found sensor name:imx415-30fps-2lane on mipi rx csi 0, i2c addr 0x1a, config\_file:linear\_3840x2160\_raw10\_30fps\_2lane.c

[1] INFO: Found sensor name:imx415-30fps-4lane on mipi rx csi 0, i2c addr 0x1a, config\_file:linear\_3840x2160\_raw10\_30fps\_4lane.c

WARN: Sensor Name: sc202cs-1600x1200, Expected Chip ID: 0xEB52, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc202cs-1600x1200, Expected Chip ID: 0xEB52, Actual Chip ID Read: 0x00

WARN: Sensor Name: irs2381c-tof, Expected Chip ID: 0x2381, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc035hgs-vc0, Expected Chip ID: 0x35, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc035hgs-vc1, Expected Chip ID: 0x35, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc231ai-30fps, Expected Chip ID: 0xCB6A, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc231ai-30fps, Expected Chip ID: 0xCB6A, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc231ai-30fps, Expected Chip ID: 0xCB6A, Actual Chip ID Read: 0x00

WARN: Sensor Name: imx586-30fps-4lane, Expected Chip ID: 0x586, Actual Chip ID Read: 0x00

WARN: Sensor Name: imx586-30fps-4lane, Expected Chip ID: 0x586, Actual Chip ID Read: 0x00

WARN: Sensor Name: os08c10-30fps-2lane, Expected Chip ID: 0x53, Actual Chip ID Read: 0x00

WARN: Sensor Name: os08c10-30fps-2lane, Expected Chip ID: 0x53, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0233-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0233-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0820std-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0820std-30fps, Expected Chip ID: 0xCB34, Actual Chip ID Read: 0x00

WARN: Sensor Name: sc1336, Expected Chip ID: 0xCA3F, Actual Chip ID Read: 0x00

WARN: Sensor Name: dummy, Expected Chip ID: 0x00, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0233-30fps, Expected Chip ID: 0xA55A, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0233-30fps, Expected Chip ID: 0xA55A, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0233-30fps, Expected Chip ID: 0xA55A, Actual Chip ID Read: 0x00

WARN: Sensor Name: ar0233-30fps, Expected Chip ID: 0xA55A, Actual Chip ID Read: 0x00

WARN: Sensor Name: ov9782-200fps-2lane, Expected Chip ID: 0x9281, Actual Chip ID Read: 0x00

WARN: Sensor Name: ov9782-200fps-2lane, Expected Chip ID: 0x9281, Actual Chip ID  
Read: 0x00

WARN: Sensor Name: ov9782-120fps-2lane, Expected Chip ID: 0x9281, Actual Chip ID  
Read: 0x00

... omitted ...